

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
4	(a)		[A surface that] absorbs all <u>em</u> radiation falling on it / perfect absorber [and emitter] of em radiation / no body is a better emitter of radiation at any wavelength than a black body at the same temperature	1			1		
	(b)	(i)	Polaris [surface] temperature > Chi Pegasi [surface] temperature (1) Intensity of radiation from Polaris > Chi Pegasi [for all λ s] accept Polaris brighter than or more luminous than Chi Pegasi (1) Polaris appears 'blue-white' and Chi Pegasi appears 'red-orange' (1) Don't accept they are different colours	3			3		
		(ii)	Peak λ identified (400 nm) (1) Application of Wien's law: $\frac{2.9 \times 10^{-3}}{400 \times 10^{-9}} = [7\,250\text{ K}](1)$ Alternative: $\frac{2.9 \times 10^{-3}}{7\,250}$ seen (1) Reference to peak wavelength on graph (400 nm) (1) Alternative: Peak wavelength (400 nm) and 7 250 K used correctly (1) to confirm Wien's constant (1)	1	1		2		
		(iii)	$L = 4.05 \times 10^{-9} \times 4\pi \times (431 \times 9.46 \times 10^{15})^2$ (1) - substitution $L = 8.46 \times 10^{29} \text{ [W]}$ seen (1)	1	1		2	2	
		(iv)	$A = \frac{8.46 \times 10^{29}}{5.67 \times 10^{-8} \times (7\,250)^4}$ (1) - substitution $A = 5.40 \times 10^{21} \text{ m}^2$ (1) $4\pi r^2 = 5.40 \times 10^{21}$ $r = 2.07 \times 10^{10} \text{ [m]}$ (1) (ecf on A)		3		3	3	
			Question 4 total	6	5	0	11	5	0